Study of the Four Papers of Erwin Schrödinger (1926)

Quantization as an Eigenvalue Problem

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Abstract

In 1926 Erwin Schrödinger published a series of four seminal papers that established the foundations of wave mechanics. These works introduced the Schrödinger equation, provided a wave-based description of quantum systems, and showed how quantized energy levels arise naturally from eigenvalue problems. This short survey reviews the main ideas of these four papers, their mathematical structure, and their impact on modern quantum mechanics.

1 Introduction

The early development of quantum theory in the 1920s raised fundamental questions about the nature of microscopic systems. Models such as Bohr's atomic theory successfully explained spectral lines but relied on rules imposed on classical orbits.

Erwin Schrödinger proposed a radically different approach: instead of describing electrons as particles following classical trajectories, one could represent them by wave functions satisfying differential equations.

In 1926 he published four papers titled *Quantisierung als Eigenwertproblem* (Quantization as an Eigenvalue Problem), which laid the foundation of wave mechanics.

2 General Contribution

Schrödinger's work introduced a mathematical framework where quantum states are represented by a complex-valued wave function

$$\psi(x,t)$$

whose evolution is governed by the Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi$$

The allowed energies of a quantum system arise as eigenvalues of an associated differential operator.

This perspective connects quantum physics with spectral theory and partial differential equations.

3 Paper I: Wave Mechanics and the Hydrogen Atom

In the first paper, Schrödinger applied his new equation to the hydrogen atom.

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

where the Coulomb potential is

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

Solving this eigenvalue problem yields the quantized energy levels

$$E_n = -\frac{me^4}{2\hbar^2 n^2}$$

which perfectly reproduce the observed hydrogen spectrum.

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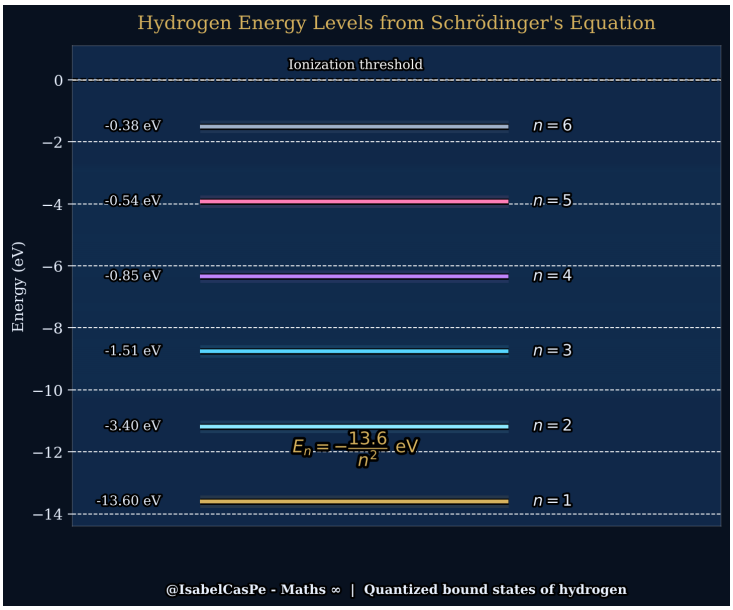

Figure 1: Hydrogen energy levels obtained from the Schrödinger equation.

4 Paper II: Connection with Classical Mechanics

The second paper explored the relationship between wave mechanics and classical mechanics.

Schrödinger used the Hamilton–Jacobi formalism to show that classical mechanics emerges as a limiting case of wave mechanics when wavelengths become very small.

The Hamilton–Jacobi equation

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}\right) = 0$$

appears as the classical limit of the Schrödinger equation.

This establishes a deep connection between wave propagation and particle trajectories.

5 Paper III: Perturbation Theory

In the third paper, Schrödinger introduced methods for systems that cannot be solved exactly.

If a Hamiltonian is written as

$$H = H_0 + \lambda V$$

one can approximate the corrected energy levels as

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$$

This approach became the foundation of perturbation theory and allowed physicists to analyze effects such as the Stark effect.

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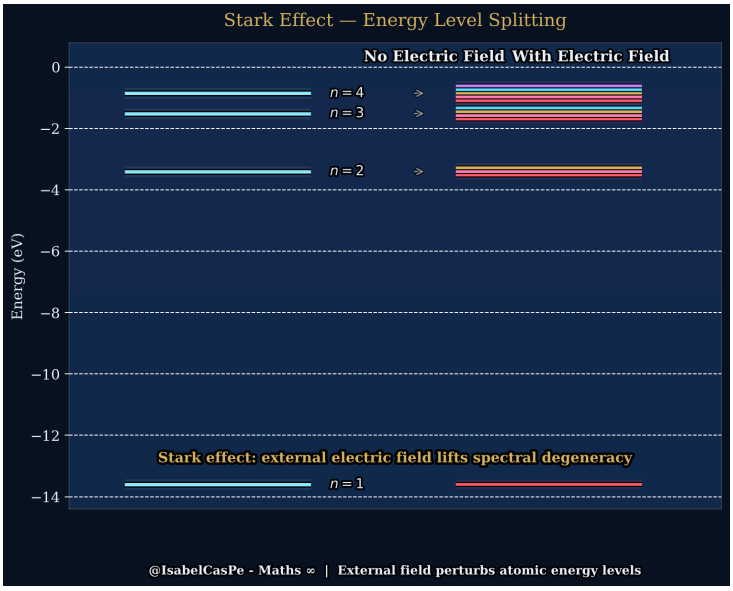

Figure 2: Energy level shifts due to an external electric field (Stark effect).

6 Paper IV: Time-Dependent Schrödinger Equation

In the fourth paper, Schrödinger introduced the time-dependent equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

This equation describes how quantum states evolve over time and allows superpositions of stationary states.

The probability density of finding a particle at position x is

$$|\psi(x,t)|^2$$

an interpretation later clarified by Max Born.

Wave packet snapshots

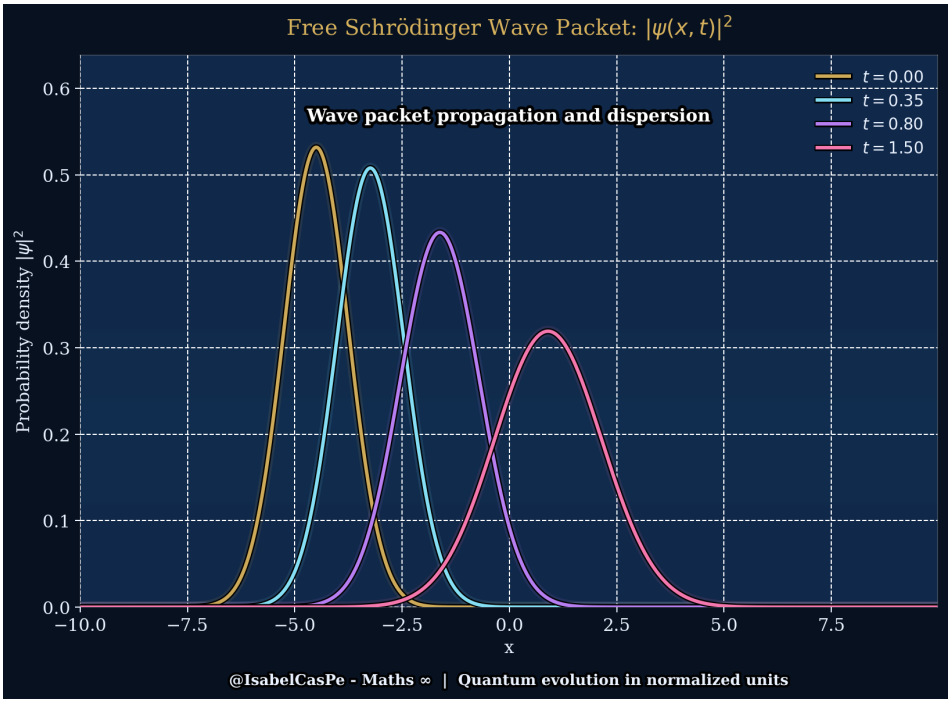

Figure 3: Snapshots of the probability density $|\psi(x,t)|^2$ for a free Gaussian wave packet.

To illustrate the dynamical interpretation of the Schrödinger equation, consider a Gaussian wave packet as an initial condition

$$\psi(x,0) = \frac{1}{(2\pi\sigma_0^2)^{1/4}} \exp\left(-\frac{(x-x_0)^2}{4\sigma_0^2}\right) \exp(ik_0 x).$$

The time evolution of the wave function is governed by the time–dependent Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}.$$

The observable quantity is the probability density

$$\rho(x,t) = |\psi(x,t)|^2.$$

Figure 3 shows several snapshots of the probability density for different times. As the system evolves, the wave packet propagates while simultaneously spreading due to dispersion.

Space-time density map

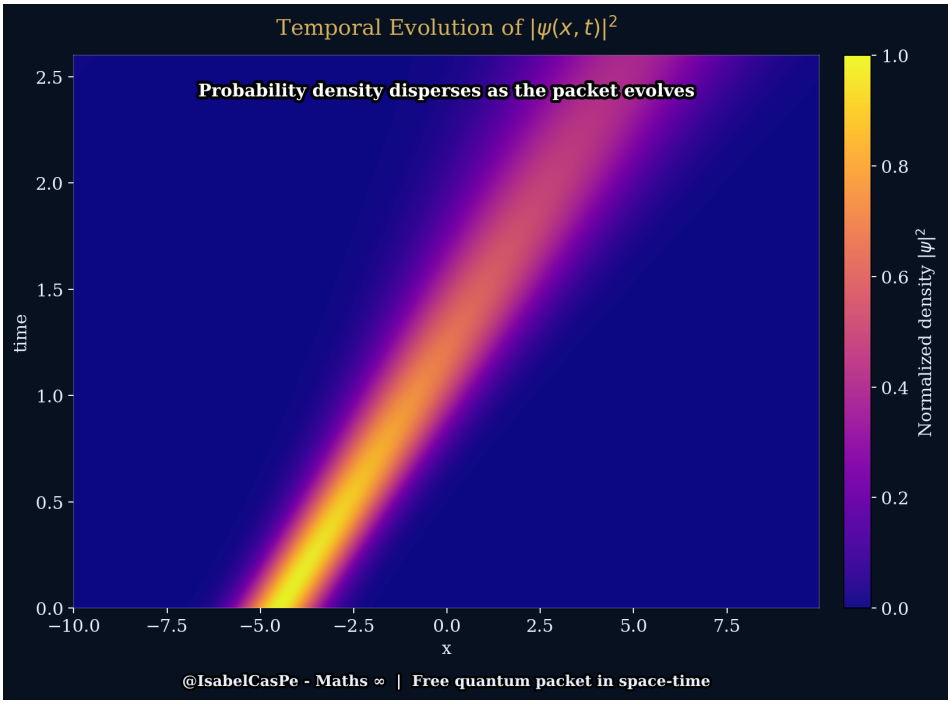

Figure 4: Space-time evolution of the probability density for the free Schrödinger equation.

Another useful way to visualize quantum dynamics is to represent the probability density in the space–time plane.

Starting from the same initial wave packet, the Schrödinger equation produces a solution

$$\psi(x,t) = \mathcal{F}^{-1}\left[\hat{\psi}_0(k)e^{-i\frac{k^2}{2}t}\right],$$

where $\mathcal{F}^{-1}$ denotes the inverse Fourier transform.

The probability density

$$\rho(x,t) = |\psi(x,t)|^2$$

can then be plotted as a function of both position and time.

Figure 4 presents a density map illustrating how the wave packet travels and disperses as time increases.

7 Conclusion

The four papers published by Schrödinger in 1926 transformed quantum theory by introducing wave mechanics as a rigorous mathematical framework.

His formulation connected quantum physics with eigenvalue problems, spectral theory, and partial differential equations.

Together with the later probabilistic interpretation proposed by Born, Schrödinger's equation became one of the central pillars of modern physics.

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